# Hidden risk in an automotive supplier base

**A proof of concept on Sayari's API for a carmaker's procurement and compliance team** · Conner Giovenco · November 2026

## The client's question

"We buy from 49 tier-1 suppliers. None of them is sanctioned. Are any of them still doing business with parties that our US, EU or UK sanctions obligations cover, through who owns them, who they own, or who they trade with?"

## What Sayari shows

**Short answer: four sets of shipments, all from German companies, arrived after a sanction took effect.** Not necessarily illegal: sanctions usually allow a *wind-down* (a grace period for older contracts), so each case needs contract dates. Strongest first:

1. **ZF → KAMAZ (Russian truck maker): after every grace period.** EU ban on 16 Mar 2022 (grace period to 15 May), asset freeze on 3 Jun. Four shipments arrived after the ban; the last (2.8 t of gearboxes, 16 Jun 2022) after both.
2. **Meritor (Cummins) → Ozkaya (Turkey): no grace period applied.** Two shipments of axle parts from Meritor Germany arrived on 5 Jan 2024, 24 days after the US sanctioned Ozkaya (SDN list: assets frozen, dealings banned).
3. **Schaeffler → AvtoVAZ (Russian carmaker): inside the US grace period.** Three shipments in Oct–Nov 2023, after the US sanctioned AvtoVAZ (14 Sep) but before its grace period ended (13 Dec); the EU listed AvtoVAZ only in 2025.
4. **Mahle → KAMAZ: possibly inside the EU grace period.** One shipment from its subsidiary Mahle Behr, 7 Apr 2022.

**Early warnings: 7 suppliers (including ZF and Schaeffler) traded directly after the invasion (24 Feb 2022) with companies the US, EU or UK sanctioned only later.** Their 13 other trades all came before the sanction: not breaches, but a sanctions-list check of these seven names finds nothing. (Bans on specific goods were not checked.)

| Supplier | Sayari's shipments to/from (count) | Latest shipment | First trade restriction |
|---|---|---|---|
| **ZF** | **to KAMAZ** (147), Russian truck maker | **16 Jun 2022** | **EU ban 16 Mar 2022: 4 after** |
| | **to Metrovagonmash** (94), Russian rail-car maker | 2 Mar 2022 | US 19 May 2023 |
| | **to AvtoVAZ** (35), Russian carmaker | 8 May 2022 | US 14 Sep 2023 |
| | **to/from ZF Kama** (5,000+), ZF–KAMAZ joint venture, Russia | 13 Dec 2022 | EU asset freeze 12 Oct 2026 |
| **Schaeffler** | **to AvtoVAZ** (313) | **7 Nov 2023** | **US 14 Sep 2023: 3 after** |
| | **to UAZ** (1), Russian vehicle maker | 30 Mar 2022 | EU asset freeze 3 Jun 2022 |
| | **to Ulyanovsk Machine Tool Plant** (14), Russia[1] | 10 Mar 2022 | EU asset freeze 18 Jul 2025 |
| | Vitesco Mexico **from Yangzhou Yangjie** (2), China[2] | 18 Jun 2024 | EU asset freeze 23 Apr 2026 |
| **Bosch** | **to ZapchastTrade** (967), Kazakhstan | 3 Sep 2024 | UK 7 Nov 2024 |
| | **from Robert Bosch Samara** (2,260), Russia[3] | 9 Feb 2023 | US 23 Feb 2024 |
| **Brose** | **to NAMI** (1), Russian state auto institute | 20 Jan 2023 | US 23 Feb 2024 |
| | **to Aurus** (1), Russian vehicle maker | 20 Apr 2022 | US 23 Feb 2024 |
| **BASF** | surfactants **to İlmor Kimya** (1), Turkey | 28 Oct 2023 | EU export restriction 20 Jul 2025 |
| **Infineon** | **to Compel** (63), Russian electronics distributor | 1 Mar 2022 | US 20 Jul 2023 |
| **Hyundai Wia** | **to ATM Grupp** (25), Russian machine-tool firm | 2 Jul 2022 | US 23 Feb 2024 |

Customs dates (dispatch, else arrival). US = OFAC SDN list; UK = UK sanctions list. Listing dates are Sayari's, except KAMAZ's EU ban; EU dates as in the Official Journal. [1] Matched by name. [2] Vitesco joined Schaeffler in Oct 2024. [3] Now NAMI Innovative Components (state-owned NAMI bought it in June 2023): a name search for "Bosch" misses it; Sayari's record links the two names.

**Next steps for the compliance team:** (1) ask ZF and Mahle for the contract dates and any authorisation behind their KAMAZ shipments, Cummins/Meritor about Ozkaya, and Schaeffler about any US link to its late-2023 AvtoVAZ shipments; (2) check whether the other relationships continued after each listing. Listing dates (Sayari's and official) for the 40 trades: `data/designation_dates.tsv`.

**All 49 suppliers at a glance** (the four cases above come first, whatever their level). A *hop* = one step (owner, subsidiary, buyer or seller); *listed* = on a US, EU or UK list today.

| Level | Suppliers | Why |
|---|---|---|
| **High** | **7** | Direct trade after 24 Feb 2022 with a now-listed party, mostly pre-listing: an early warning, not a breach |
| **Medium** | **26** | **22** linked at two hops after the invasion (incl. Mahle, Cummins and Meritor); **1** traded directly only before it (JTEKT); **3** with only a forced-labor flag (Adient, ThyssenKrupp, Toyoda Gosei) |
| **Low** | **16** | No rule triggered; for **5** the matched Sayari records hold little data, so "low" means "unknown," not "clean" |

In all, **30 of 49 suppliers** appear in the appendix. Most two-hop chains run through another company's trade (weak evidence); stronger ones run through **the supplier's own subsidiary**: at least 10 suppliers have one that traded with a now-listed party after the invasion (25 trades: 20 before the listing; Mahle's and Meritor's after; 2 to YaMZ, whose parent GAZ has been US-sanctioned since 2018; 1 probable false match). One ownership chain Sayari shows as current also counts: BASF → Wintershall Dea → Achim Development.

## Approach

1. **Match** each supplier to every Sayari record with its exact legal name (`/v1/resolution`, `/v1/entity_summary`).
2. **Pull** risk flags, ownership and trade paths (not ended) within two hops to listed parties (`/v1/watchlist`), individual shipment dates for direct and subsidiary trades (`/v1/trade/search/shipments`), and direct shippers (`/v1/supply_chain/upstream`).
3. **Score** with my own triage rules (screening, not a legal finding), auditable by a compliance officer: **high** = a current direct ownership link, or direct trade dated after 24 Feb 2022, with a party now on a US/EU/UK list; **medium** = an older or undated direct link, a two-hop link since then or of unknown date, or a high/critical export-control flag within 2 hops or forced-labor flag within 3; **low** = none of these.
4. **Date each listing** with Sayari's listing dates, checked against official sources for the 40 post-invasion trades: all match except KAMAZ's EU ban (16 Mar 2022), which Sayari misses.

## Assumptions

- Only **US, EU and UK** sanctions and export-control lists count (the client operates there). A link must run **one way** (up or down ownership, or along trade) or go **down to a subsidiary and then trade**; shared investors, suppliers or customers don't count. A chain is as recent as its **least recent trade link** (each link dated by its latest shipment); only **high/critical** flags count. Full rules: README.

## Challenges and how they were handled

- **Matching:** Sayari's top "strong" match was sometimes the wrong company (Bosch → Bosch Vietnam). Legal names plus a legal-form check fixed it; the 19 weaker matches were **checked by hand** (2 corrected; see README).
- **The first scoring overstated risk:** 14 "high" suppliers, cut to 3 by three compliance questions: *Is that really their owner?* (shared investors removed); *Is that list binding on us?* (Chinese/Russian counter-sanctions removed); *When did it happen?* (trade before 24 Feb 2022 separated).
- **Dates changed the headline three times.** One of BASF's split records held the İlmor shipment (4 high). Dating trades by their latest shipment, not the first, added ZF, Infineon and Hyundai Wia (7). Then the shipment search, which dates every shipment, showed what the summary missed: ZF's KAMAZ deliveries ran to June 2022, Schaeffler's AvtoVAZ ones to Nov 2023, Meritor's Ozkaya ones to Jan 2024.

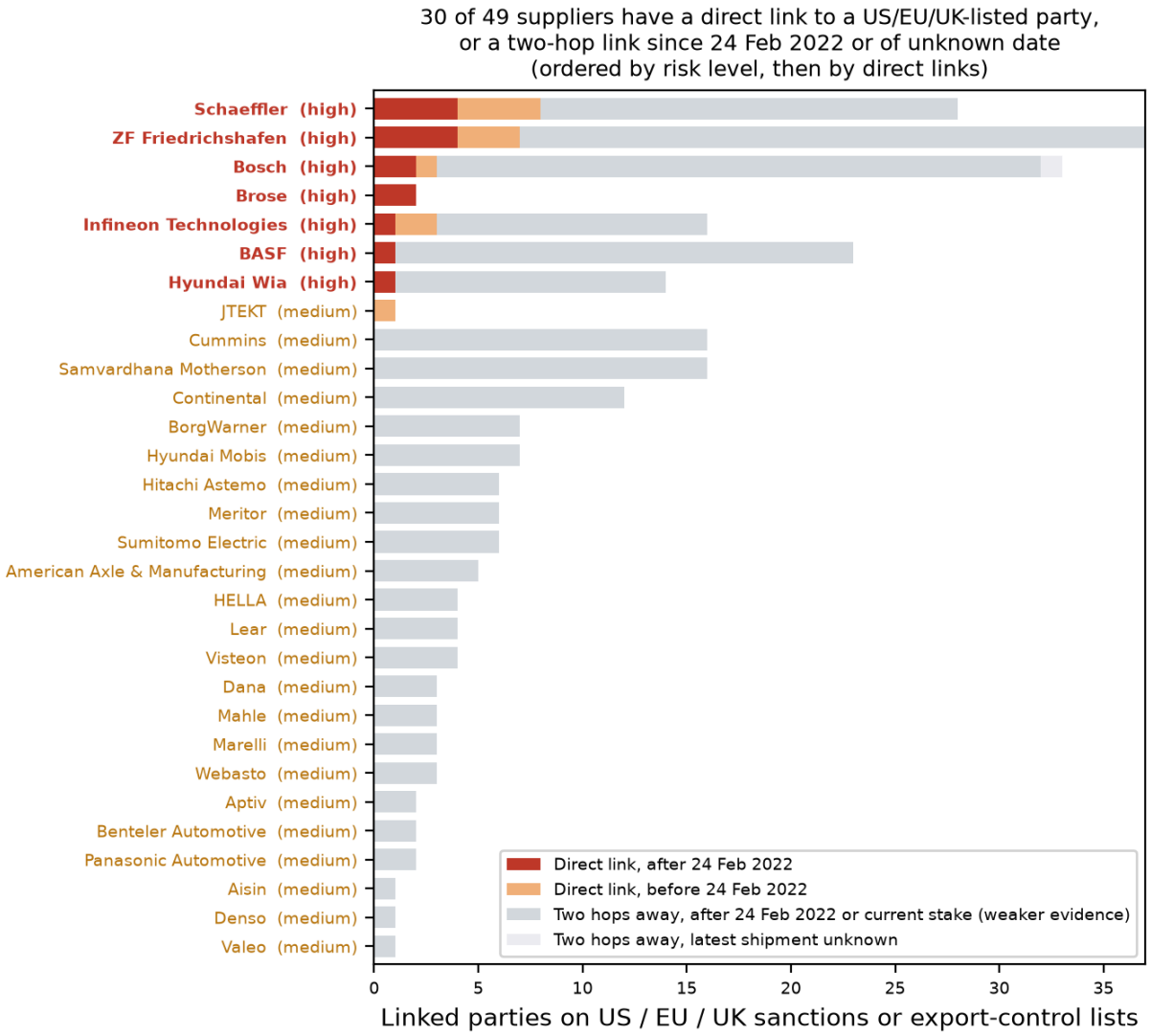
## Why this needs Sayari

A name screen of these 49 suppliers finds nothing. Sayari's shipment data, ownership graph and per-list tags found ZF's gearboxes reaching KAMAZ after the EU ban, post-restriction shipments by two subsidiaries, a renamed sanctioned supplier that name searches miss (Robert Bosch Samara), and 7 suppliers trading with firms before regulators acted.

## What I'd do next

- **Score on Sayari's listing dates** (already in the output), not the invasion date, **monitor** with Sayari's alerts, and **match by ID** (LEI or tax number; Sayari holds one for 46 of the 49) from the client's vendor master.
- **Go deeper:** tier-2/3 suppliers, negative news, labor risk (5 suppliers have a shipper flagged for forced labor).

## Appendix: the 30 suppliers with a direct or recent link




A long grey bar (many two-hop links) is weaker evidence than one red segment. One of Bosch's two post-invasion direct links (red) is Robert Bosch Samara (SDN-listed 23 Feb 2024 under its new name, NAMI Innovative Components; Bosch's last shipment was Feb 2023). Full detail per supplier: `output/supplier_risk.csv`. Every linked party with its closest path, date (and whether the date is observed or a bound), lists and Sayari ID: `output/watchlist_links.csv`. Match evidence: `output/matches.csv`.